Each side label is a SQUARED vector length

Orthogonal triangles in observation space; schematic, not to scale.

A GENERAL NESTED MODELS

Reduced fitted values can differ across observations.

$$SSE_R = (SSE_R - SSE_F) + SSE_F$$

B REDUCED MODEL: INTERCEPT ONLY

$$\hat{\mathbf{Y}}_R = \left(\overline{Y}, \ldots, \overline{Y}\right)^T$$

$$SST = SSR + SSE$$

The GREEN side joins the two fitted vectors; its squared length is the extra SS.